

BigQuery Star Schema Data Model

Jaybel Sales Analytics Documentation

Project: jaybel-dev | Dataset: jaybel_sales_analytics

1. Overview

This document describes the BigQuery data warehouse for Jaybel Sales Analytics. Data was migrated from AWS RDS MySQL and modelled using a Star Schema for efficient analytical queries and Power BI reporting.

Layer	Description
Staging (stg_*)	Raw tables from RDS MySQL. All columns STRING. No transforms applied.
Dimensions (dim_*)	Cleaned typed lookup tables: date, customer, sales rep, product, department.
Facts (fact_*)	Transactional tables with FK references to all dimensions and typed numeric measures.

Fiscal Year Convention: July = Fiscal Month 1, June = Fiscal Month 12. Label format: 2025-2026 (not FY2026). Date range in dim_date: 2018-07-01 to 2030-06-30.

2. Staging Tables (stg_*)

All staging columns are STRING type. Type casting is applied during star schema transforms.

1.)stg_sales_report

Source: sales_file_details (RDS MySQL)

Raw sales transaction data. Each row = one invoice line item. All columns are STRING.

Column Name	Data Type	Description
id	STRING	Unique record identifier
account	STRING	Customer account code
trans_date	STRING	Transaction date (datetime string)
invoice	STRING	Invoice number
order_no	STRING	Order number
group_no	STRING	Product group number
item_code	STRING	Product item code
brand_manuf	STRING	Brand or manufacturer name
manuf_apn_no	STRING	Manufacturer APN number
description	STRING	Product description
qty	STRING	Quantity sold
line_sales_ex_gst	STRING	Line total sales excluding GST
line_cost	STRING	Line total cost
unit_sale_ex_gst	STRING	Unit selling price ex GST
unit_cost	STRING	Unit cost price
unit_gp_dollar	STRING	Unit gross profit in dollars
line_gp_dollar	STRING	Line gross profit in dollars
gp_percent	STRING	Gross profit percentage
rep	STRING	Sales rep code
mkt	STRING	Market segment code
account_name	STRING	Customer account name
whse	STRING	Warehouse code
ter	STRING	Territory code
orgn_rep	STRING	Original rep code at time of sale
group_name	STRING	Product group name
main_group_name	STRING	Main product group name
rep_name	STRING	Sales rep full name

Column Name	Data Type	Description
company	STRING	Company name: Jaybel or Frazer
fiscal_year	STRING	Fiscal year label e.g. 2025-2026
territory	STRING	Territory name
frazer_link	STRING	Frazer system link identifier
file_id	STRING	Source file ID for traceability
industry	STRING	Industry classification
industry_sub	STRING	Industry sub-classification
business_code	STRING	Business type code

2.)stg_sales_cust

Source: sales_customer (RDS MySQL)

Raw customer account data.

Column Name	Data Type	Description
id	STRING	Unique customer identifier
account	STRING	Customer account code (join key)
account_name	STRING	Customer account name
created_date	STRING	Date account was created
first_created_date	STRING	First transaction date for account
rep	STRING	Assigned sales rep code
is_new_account	STRING	Flag: 1=new account, 0=existing
is_new_created_date_cycle	STRING	Flag: 1=new in current cycle, 0=not
industry_sub	STRING	Industry sub-classification
business_code	STRING	Business type code

3.)stg_sales_rep

Source: sales_rep (RDS MySQL)

Raw sales representative reference data.

Column Name	Data Type	Description
sales_rep_id	STRING	Unique sales rep identifier
sales_rep_code	STRING	Short rep code used in transactions
description	STRING	Full name or description of rep
frazer_link	STRING	Frazer system link identifier

Column Name	Data Type	Description
company	STRING	Company: Jaybel or Frazer

4.)stg_main_product

Source: power_bi_main_product_group (RDS MySQL)

Raw product group reference data.

Column Name	Data Type	Description
id	STRING	Unique product group identifier
main_group_name	STRING	Name of the main product group

5.)stg_new_busin_frazer

Source: new_busin_fraz (RDS MySQL)

Raw new business transactions for Frazer. Kept separate from main sales.

Column Name	Data Type	Description
account	STRING	Customer account code
account_name	STRING	Customer account name
trans_date	STRING	Transaction date (datetime string)
rep	STRING	Sales rep code
group_no	STRING	Product group number
main_group_name	STRING	Main product group name
line_sales_ex_gst	STRING	Line sales amount excluding GST
line_gp_dollar	STRING	Line gross profit in dollars
company	STRING	Company name (Frazer)
territory	STRING	Territory name
frazer_link	STRING	Frazer system link identifier

6.)stg_total_working_days

Source: total_working_days (RDS MySQL)

Standalone reference table for working days per fiscal month. Not part of the star schema.

Column Name	Data Type	Description
id	STRING	Unique record identifier

Column Name	Data Type	Description
fiscal_year	STRING	Fiscal year label e.g. 2025-2026
month	STRING	Month name or number
total_working_days	STRING	Total working days in that month
YearlyTotalWorkingDays	STRING	Total working days for the full fiscal year
Completed_days	STRING	Number of working days completed so far

3. Dimension Tables (dim_*)

Dimension tables provide descriptive attributes for facts. Bold column names are PKs or FKs.

7.)dim_date

Date dimension generated programmatically. Fiscal year: July=Month 1, June=Month 12. e.g. July 2025 to June 2026 = FY 2025-2026. Date range: 2018-07-01 to 2030-06-30.

Column Name	Data Type	Description
date_key	INT64	Surrogate key YYYYMMDD format (PK)
date	DATE	Calendar date
day_name	STRING	Full day name e.g. Monday
day_name_short	STRING	Short day name e.g. Mon
day_of_month	INT64	Day number in month (1-31)
calendar_month_no	INT64	Calendar month number (1-12)
calendar_month_name	STRING	Calendar month name e.g. January
calendar_year	INT64	Calendar year e.g. 2025
calendar_quarter	INT64	Calendar quarter (1-4)
fiscal_month_no	INT64	Fiscal month number (July=1, June=12)
fiscal_month_name	STRING	Fiscal month name e.g. July
fiscal_year	INT64	Fiscal year as integer e.g. 2026
fy	STRING	Fiscal year label e.g. 2025-2026
fiscal_quarter	STRING	Fiscal quarter: Q1=Jul-Sep, Q2=Oct-Dec, Q3=Jan-Mar, Q4=Apr-Jun
is_weekday	BOOL	True if Monday to Friday, False if weekend

8.)dim_department

Department dimension with exactly two entries: Jaybel (key=1) and Frazer (key=2). Created manually.

Column Name	Data Type	Description
department_key	INT64	Surrogate key (PK): 1=Jaybel, 2=Frazer
department_name	STRING	Department name: Jaybel or Frazer

9.)dim_sales_customer

Customer dimension sourced from stg_sales_cust. Each row is a unique customer account.

Column Name	Data Type	Description
customer_key	INT64	Surrogate key (PK)
account	STRING	Customer account code (natural key)
account_name	STRING	Customer account name
created_date	DATE	Date account was created
first_created_date	DATE	First transaction date for account
rep	STRING	Assigned sales rep code
is_new_account	BOOL	True if new account in current period
is_new_created_date_cycle	BOOL	True if new in current fiscal cycle
industry_sub	STRING	Industry sub-classification
business_code	STRING	Business type code

10.)dim_sales_rep

Sales representative dimension sourced from stg_sales_rep.

Column Name	Data Type	Description
rep_key	INT64	Surrogate key (PK)
sales_rep_id	STRING	Original rep ID from source system
sales_rep_code	STRING	Short rep code used in transactions (natural key)
description	STRING	Full name or description of the rep
frazier_link	STRING	Frazier system link identifier
company	STRING	Company the rep belongs to
department_key	INT64	FK to dim_department

11.)dim_product

Product group dimension sourced from stg_main_product.

Column Name	Data Type	Description
product_key	INT64	Surrogate key (PK)
main_group_name	STRING	Main product group name (natural key)

4. Fact Tables (fact_*)

Fact tables store transactional measures and FK references to all five dimensions. Measures are FLOAT64.

12.)fact_sales_report

Primary fact table. Grain: one row per invoice line item. Sourced from stg_sales_report.

Column Name	Data Type	Description
sales_key	INT64	Surrogate key (PK)
date_key	INT64	FK to dim_date
customer_key	INT64	FK to dim_customer
rep_key	INT64	FK to dim_sales_rep
product_key	INT64	FK to dim_product
department_key	INT64	FK to dim_department
invoice	STRING	Invoice number
order_no	STRING	Order number
item_code	STRING	Product item code
whse	STRING	Warehouse code
territory_code	STRING	Territory code
orgn_rep	STRING	Original rep code at time of sale
mkt	STRING	Market segment code
file_id	STRING	Source file ID for traceability
qty	FLOAT64	Quantity sold
line_sales_ex_gst	FLOAT64	Line total revenue excluding GST
line_cost	FLOAT64	Line total cost
line_gp_dollar	FLOAT64	Line gross profit in dollars
gp_percent	FLOAT64	Gross profit percentage
unit_sale_ex_gst	FLOAT64	Unit selling price ex GST
unit_cost	FLOAT64	Unit cost price
unit_gp_dollar	FLOAT64	Unit gross profit in dollars

13.)fact_new_business

Separate fact table for new business transactions from Frazer. Sourced from stg_new_busin_frazer. Shares all five dimensions with fact_sales.

Column Name	Data Type	Description
new_business_key	INT64	Surrogate key (PK)
date_key	INT64	FK to dim_date
customer_key	INT64	FK to dim_customer
rep_key	INT64	FK to dim_sales_rep
product_key	INT64	FK to dim_product
department_key	INT64	FK to dim_department
territory	STRING	Territory name
frazer_link	STRING	Frazer system link identifier
line_sales_ex_gst	FLOAT64	Line sales amount excluding GST
line_gp_dollar	FLOAT64	Line gross profit in dollars

5. Foreign Key Relationships

Both fact tables share the same five dimensions via these join keys:

Fact Column	Joins To	Dimension Table	Join Condition
date_key	date_key	dim_date	DATE(TIMESTAMP(trans_date)) = date
customer_key	customer_key	dim_customer	TRIM(account) = TRIM(account)
rep_key	rep_key	dim_sales_rep	TRIM(rep) = TRIM(sales_rep_code)
product_key	product_key	dim_product	TRIM(main_group_name) = TRIM(main_group_name)
department_key	department_key	dim_department	LOWER(TRIM(company)) = LOWER(TRIM(department_name))